



NHL Zone Entries – Controlled vs Dump-and-Chase, which is more effective?

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The Question



Controlled Zone Entries:

- One player “carries” the puck into the offensive zone while maintaining possession

Dump-and-Chase Zone Entries:

- One player dumps the puck into the offensive zone, initiating a battle to regain possession

Which of these entry strategies, over the course of a season, generates more offensive production at the team-level?

Model



$$\text{xGF60}_{it} = \beta_0 + \beta_1 \text{ControlledEntryRate}_{it} + \beta_2 \text{FF60}_{it} + \beta_3 \text{CF60}_{it} + \beta_4 \text{ShootingPct}_{it} + \beta_5 \text{PPPct}_{it} + \beta_6 \text{PointsPct}_{it} + \gamma_t + \epsilon_{it}$$

where:

- i = team
- t = season
- γ_t = season fixed effects
- ϵ_{it} = error term
- **xGF60** = expected goals for per 60 minutes
- **ControlledEntryRate** = total Controlled Entries / total Entries
- **FF60** = unblocked shot attempts per 60 minutes
- **CF60** = all shot attempts (block OR unblocked) per 60 minutes
- **ShootingPct** = (total goals scored / total shots on goal) x 100
- **PPPct** = team powerplay% in season t
- **PointsPct** = team standings points% in season t

Data sources: All Three Zones, Moneypuck, Evolving Hockey, ESPN

*Zone entry data was only available for **three seasons (2022-25)** – 99 team-season observations*

Model Results



OLS Regression Results: Factors Affecting Expected Goals For (xGF/60)		
Term	Estimate	p-value
Intercept	-0.899**	0.006
Controlled Entry Rate	0.481	0.197
Fenwick For (FF/60)	0.047*	0.014
Corsi For (CF/60)	0.02	0.135
Shooting Percentage	4.976*	0.019
Power Play Percentage	1.534***	<0.001
Points Percentage	-0.243	0.267
R ² = 0.842 Adjusted R ² = 0.828 Observations (N) = 99 Stars: *** p<0.001, ** p<0.01, * p<0.05		

Model Diagnostics and Fit



Model Assumption Checks			
Analysis of Residuals for Zone Entry Model			
Test	Statistic	P-Value	Interpretation
Heteroscedasticity (Breusch-Pagan)	10.33	0.242	Constant Variance (Good)
Autocorrelation (Durbin-Watson)	1.76	0.075	No Autocorrelation

$$R^2 = 0.842$$

- Model meets those OLS assumptions
 - Results are robust and reliable

Model Interpretations/Analysis



Fenwick For Per 60

Term	Estimate	p-value
Fenwick For Per 60	0.047*	0.014

- Might seem like a small estimate, however it is not
- 5 extra unblocked shots per game leads to about 19 additional xG over a season

Shooting Percentage

Term	Estimate	p-value
Shooting Percentage	4.976*	0.019

- Better shooting team = more talent = more xGF, as expected

Powerplay Percentage

Term	Estimate	p-value
Powerplay Percentage	1.534***	<0.001

- Very significant p-value also as expected, more xGF occur when on the powerplay

Model Interpretations/Analysis



Controlled Entry Rate is the KEY variable when answering the question

Ho: there is no relationship between a teams controlled entry rate and their xGF60

Ha: there is a relationship between a teams controlled entry rate and their xGF60



Controlled Entry Rate		
Term	Estimate	p-value
Controlled Entry Rate	0.481	0.197

- Fail to reject Ho - Can not conclude that there is a relationship between a teams controlled entry rate and their xGF60

What does this mean?

- Teams that primarily carry the puck into the zone do not statistically generate more expected goals.

Takeaways/Answering the Question



“Which of these entry strategies, over the course of a season, generates more offensive production at the team-level?”

Entry strategy is flexible:

- No statistical evidence that one strategy is “better” than the other
- Adapt based on the defense